Unit-1

Q.1 Define computer network & List out applications of Computer Network

A computer network is a system that connects numerous independent computers in order to share information (data) and resources. The integration of computers and other different devices allows users to communicate more easily.

Some of the network applications in different field are the following :

- **Marketing and sales: –** Computer networks are widely used in both marketing sales firms. These are used by marketing professionals to collect, exchange, and analyzes data relating to customer requirements and product development cycles. Teleshopping is also important part of sales applications that use order-entry computers or telephones connected to an order-processing network, and on-line reservation services for hotels airline and so on.

- **Manufacturing:** – Now days, computer networks are used in a several aspects of manufacturing, including the manufacturing process itself. Two applications which use a network to provide necessary services are computer-assisted manufacturing (CAM) and computer –assisted designing (CAD) both of which permit multiple users to work on a project simultaneously.
- **Financial Services: –** In Present, Financial services are completely dependent on computer networks. Main applications are credit history searches, foreign exchange and investment services, and Electronic Funds Transfer (EFT) that permits a user to transfer money without going into bank.
- **Teleconferencing:** – With The help of teleconferencing conferences are possible to occur without the participants being in the same place. Applications include simple text conferencing, voice conferencing, and video conferencing.
- **Cable Television:**-Future Services provided by cable television network can include video on request, as well as the same information, financial and communications services currently provided by the telephone companies and computer networks.
- **Information Services:-** Network information services include bulletin boards and data banks. A World Wide Web site offering the technical specifications for a new product is an information service.

- **Electronic Messaging:**– Electronic mail (e-mail) is the most widely used network application.
- **Electronic Data Interchange (EDI):**– EDI permits business information to be transferred without using paper.
- **Directory services:** – By using directory services, it is possible to store the last of files in a central location to speed worldwide search operations.
- **Cellular Telephone: –** In the past, two parties desiring to use the services of the telephone company had to be linked by a fixed physical connection. But, in present cellular network make it possible to maintain wireless phone connections even while travelling over large distances.

Q.2 Define LAN, MAN, WAN Define connection-oriented and connectionless Protocols.

LAN - Local Area Network

A Local Area Network (LAN) is a private network that connects computers and devices within a limited area like a residence, an office, a building or a campus. On a small scale, LANs are used to connect personal computers to printers. However, LANs can also extend to a few kilometers when used by companies, where a large number of computers share a variety of resources like hardware (e.g. printers, scanners, audiovisual devices etc), software (e.g. application programs) and data.

MAN - Metropolitan Area Network

A Metropolitan Area Network (MAN) is a larger network than LAN. It often covers multiple cities or towns. It is quiet expensive and a single organization may not have own it.

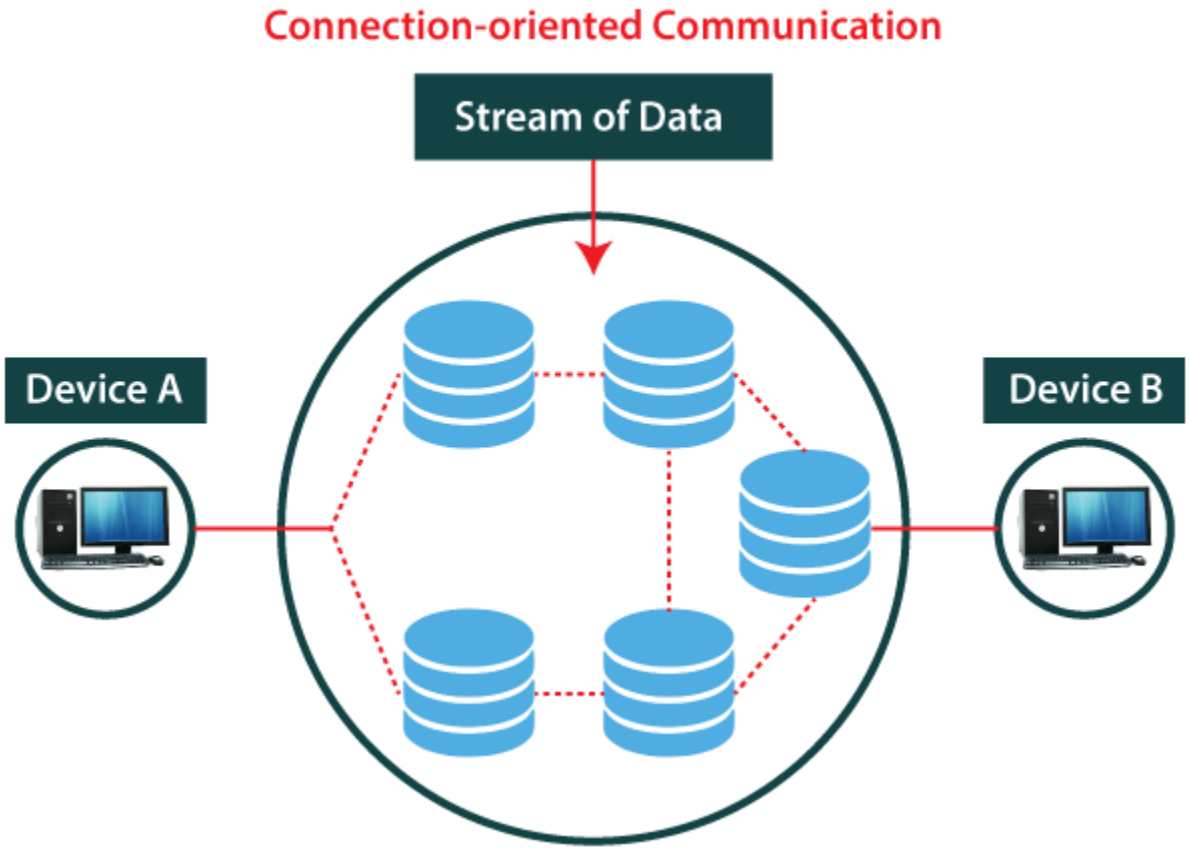
WAN - Wide Area Network

A Wide Area Network (WAN) is a much larger network than LAN and MAN. It often covers multiple contries or contenants. It is quiet expensive and a single organization may not have own it. Satellite is used to manage WAN.

Data communication is a telecommunication network to send and receive data between two or more computers over the same or different network. There are two ways to establish a connection before sending data from one device to another, that are **Connection-Oriented** and **Connectionless Service**. Connection-oriented service involves the creation and termination of the connection for sending the data between two or more devices. In contrast, connectionless service does not require establishing any connection and termination process for transferring the data over a network.

Connection-Oriented Service

A connection-oriented service is a network service that was designed and developed after the telephone system. A connection-oriented service is used to create an end to end connection between the sender and the receiver before transmitting the data over the same or different networks. In connection-oriented service, packets are transmitted to the receiver in the same order the sender has sent them. It uses a handshake method that creates a connection between the user and sender for transmitting the data over the network. Hence it is also known as a reliable network service.

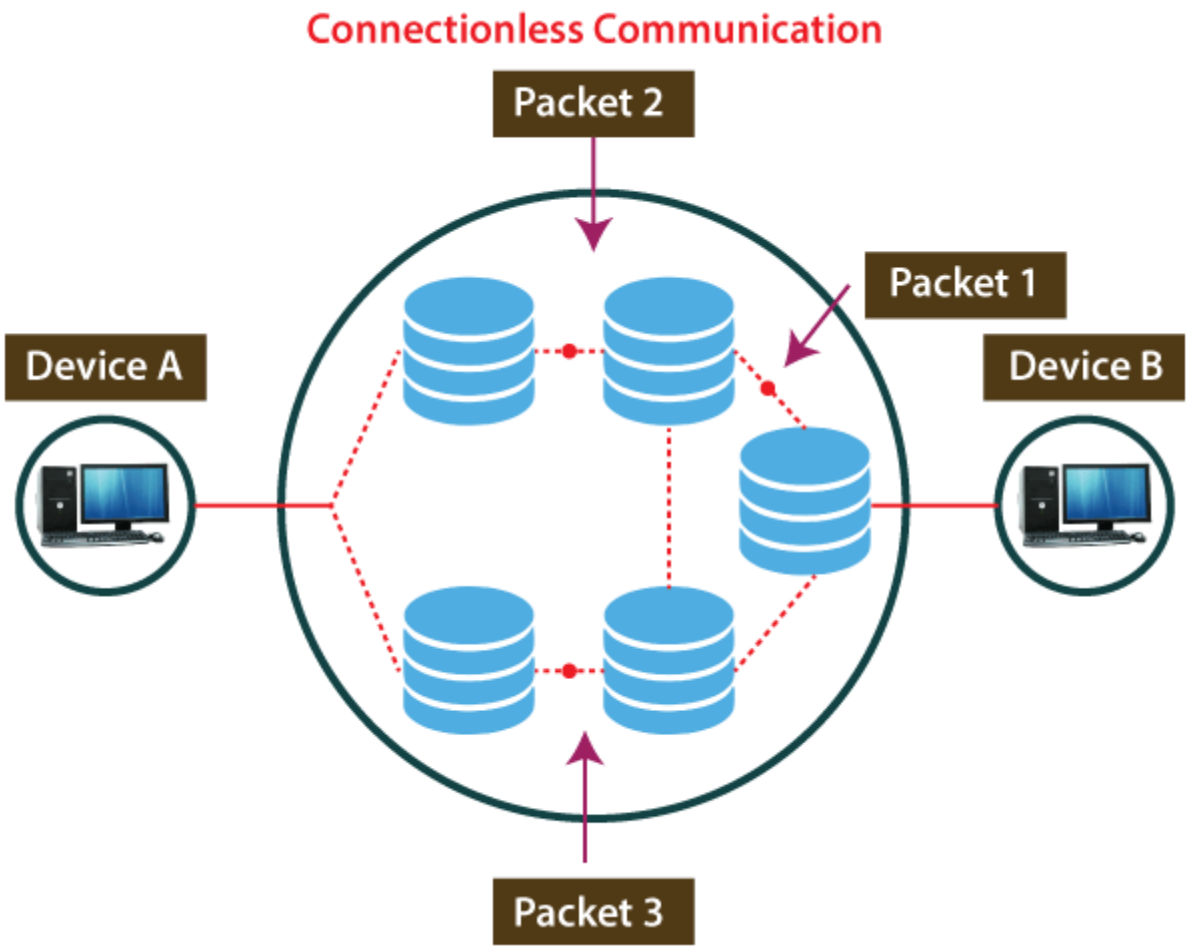


Suppose, a sender wants to send data to the receiver. Then, first, the sender sends a request packet to a receiver in the form of an **SYN** packet. After that, the receiver responds to the sender's request with an (SYN-ACK) signal/packets. That represents the confirmation is received by the receiver to start the communication between the sender and the receiver. Now a sender can send the message or data to the receiver.

Similarly, a receiver can respond or send the data to the sender in the form of packets. After successfully exchanging or transmitting data, a sender can terminate the connection by sending a signal to the receiver. In this way, we can say that it is a reliable network service.

Connectionless Service

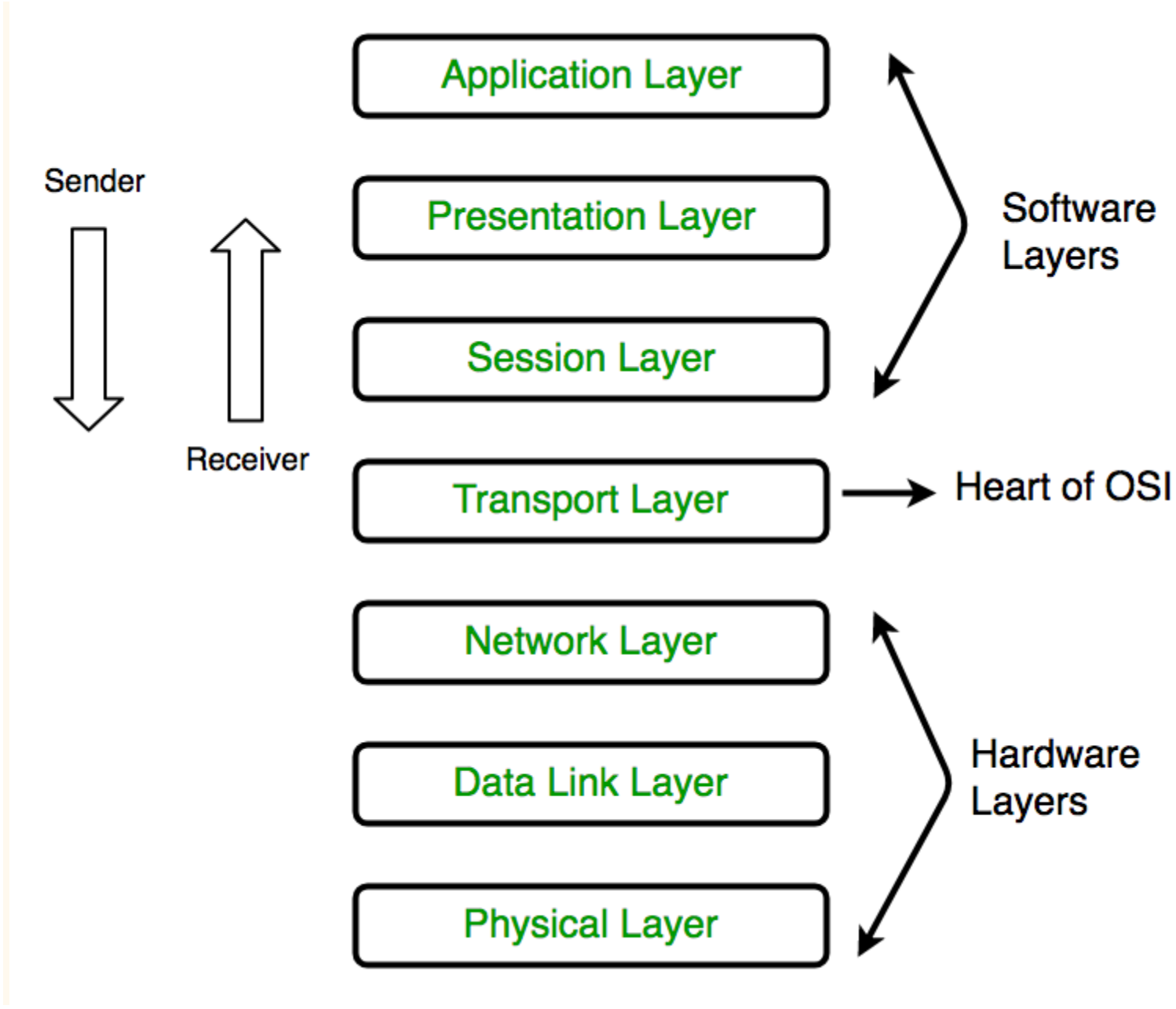
A connection is similar to a **postal system**, in which each letter takes along different route paths from the source to the destination address. Connectionless service is used in the network system to transfer data from one end to another end without creating any connection. So it does not require establishing a connection before sending the data from the sender to the receiver. It is not a reliable network service because it does not guarantee the transfer of data packets to the receiver, and data packets can be received in any order to the receiver. Therefore we can say that the data packet does not follow a **defined** path. In connectionless service, the transmitted data packet is not received by the receiver due to network congestion, and the data may be lost.



For example, a sender can directly send any data to the receiver without establishing any connection because it is a connectionless service. Data sent by the sender will be in the packet or data streams containing the receiver's address. In connectionless service, the data can be travelled and received in any order. However, it does not guarantee to transfer of the packets to the right destination.

Q.3 Draw OSI Reference Model and explain it in detail.

OSI or Open System Interconnection model was developed by International Standards Organization (ISO). It gives a layered networking framework that conceptualizes how communications should be done between heterogeneous systems. It has seven interconnected layers. The seven layers of the OSI Model are a physical layer, data link layer, network layer, transport layer, session layer, presentation layer, and application layer, as shown in the following diagram –


Application Layer
Presentation Layer
Session Layer
Transport Layer
Network Layer
Data Link Layer
Physical Layer
Sender
Receiver
Software Layers
Heart of OSI
Hardware Layers

The physical layer, data link layer and the network layer are the network support layers. The layers manage a physical transfer of data from one device to another. Session layer, presentation layer, and application layer are the user support layers. These layers allow communication among unrelated software in dissimilar environments. Transport layer links the two groups.

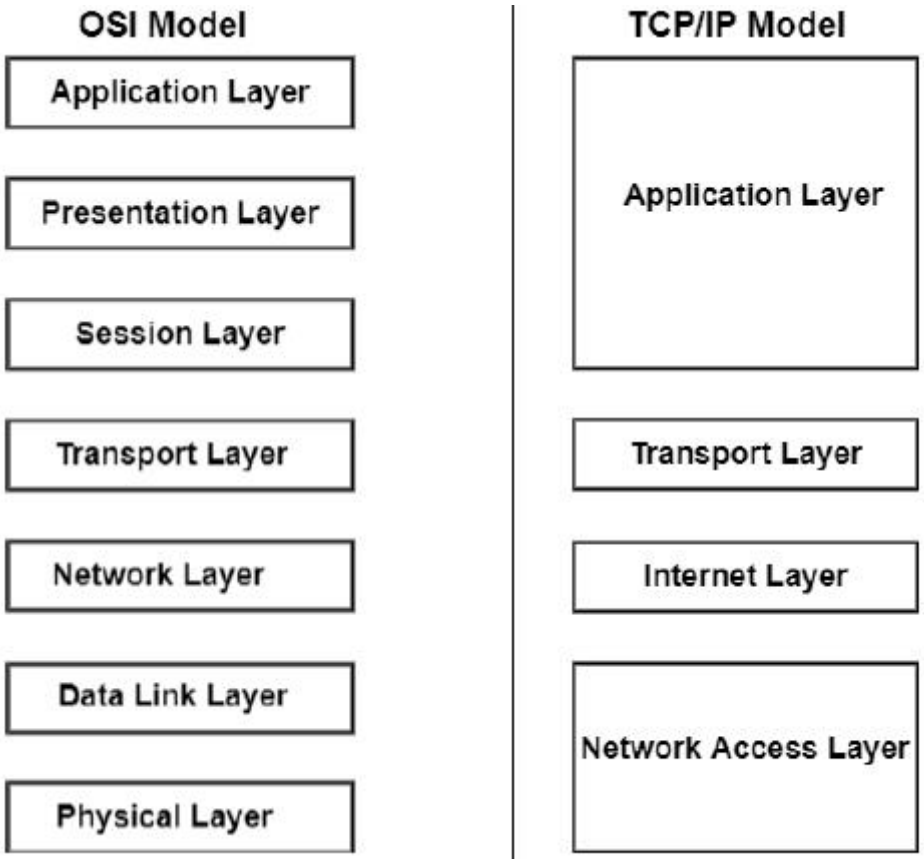
The main functions of each of the layers are as follows –

- **Physical Layer –** Its function is to transmit individual bits from one node to another over a physical medium.
- **Data Link Layer –** It is responsible for the reliable transfer of data frames from one node to another connected by the physical layer.
- **Network Layer –** It manages the delivery of individual data packets from source to destination through appropriate addressing and routing.
- **Transport Layer –**It is responsible for delivery of the entire message from the source host to destination host.
- **Session Layer –** It establishes sessions between users and offers services like dialog control and synchronization.
- **Presentation Layer –** It monitors syntax and semantics of transmitted information through translation, compression, and encryption.
- **Application Layer –** It provides high-level APIs (application program interface) to the users.

4. Compare OSI & TCP-IP reference model.

OSI represents **Open System Interconnection**.	TCP/IP model represents the Transmission Control Protocol / Internet Protocol.
OSI is a generic, protocol independent standard. It is acting as an interaction gateway between the network and the final-user.	TCP/IP model depends on standard protocols about which the computer network has created. It is a connection protocol that assigns the network of hosts over the internet.
The OSI model was developed first, and then protocols were created to fit the network architecture's needs.	The protocols were created first and then built the TCP/IP model.
It provides quality services.	It does not provide quality services.
The OSI model represents defines administration, interfaces and conventions. It describes clearly which layer provides services.	It does not mention the services, interfaces, and protocols.
The protocols of the OSI model are better unseen and can be returned with another	The TCP/IP model protocols are not hidden, and

appropriate protocol quickly.	we cannot fit a new protocol stack in it.
It is difficult as distinguished to TCP/IP.	It is simpler than OSI.
It provides both connection and connectionless oriented transmission in the network layer; however, only connection-oriented transmission in the transport layer.	It provides connectionless transmission in the network layer and supports connecting and connectionless-oriented transmission in the transport layer.
It uses a horizontal approach.	It uses a vertical approach.
The smallest size of the OSI header is 5 bytes.	The smallest size of the TCP/IP header is 20 bytes.
Protocols are unknown in the OSI model and are returned while the technology modifies.	In TCP/IP, returning protocol is not difficult.



5. Compare circuit and packet switching.

Circuit Switching

Circuit Switching is a connection-oriented service. It provides a dedicated path from the sender to the receiver. In-circuit switching, a connection setup is required to send and receive data. It has very little chance of data loss and error due to the dedicated circuit, but a lot of bandwidth is wasted because the same path cannot be used by other senders during a congestion. Circuit switching is completely transparent; the sender and receiver can use any bit rate format or framing method.

Advantages of Circuit Switching

- It uses a fixed bandwidth.
- A dedicated communication channel increases the quality of communication.
- Data is transmitted with a fixed data rate.
- No waiting time at switches.
- Suitable for long continuous communication.

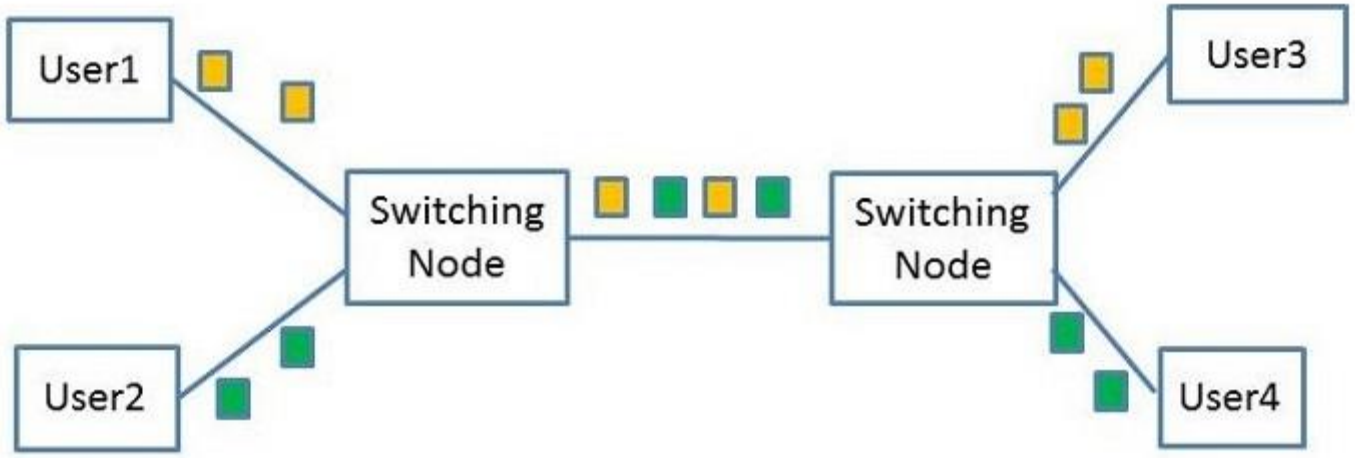
Disadvantages of circuit switching

- A dedicated connection makes it impossible to transmit other data even if the channel is free.
- Resources are not utilized fully.
- The time required to establish the physical link between the two stations is too long.

- A dedicated path has to be established for each connection.
- Circuit switching is more expensive.
- Even if there is no transfer of data, the link is still maintained until it is terminated by users.
- Dedicated channels require more bandwidth.

Packet Switching

Packet switching is a connectionless service. It does not require any dedicated path between the sender and receiver. It places an upper limit on block size. In packet switching bandwidth is freely utilized as unrelated sources can be used in any path. It has more chance of data loss and error; the packets may arrive in the wrong order.



Packet Switching

Advantages of Packet switching

- It reduces access delay.
- Costs are minimized to great extent. Hence packet switching is a very cost-effective technique.
- Packets are rerouted in case of any problems. This ensures reliable communication.
- It is more efficient for data transmission because no need to establish the path.

- Several users can share the same channel simultaneously. Therefore packet switching makes use of available bandwidth efficiently.

Disadvantages of Packet switching

- In packet switching, the network can not be used in applications requiring very little delay and higher quality of service.
- Protocols used in the packet switching are complex.
- If the network becomes overloaded, packets are delayed or discarded, or dropped. This leads to the retransmission of lost packets by the sender.
- It is not secured if security protocols are not used during packet transmission.

Difference between circuit switching and packet switching

The following table highlights the major differences between circuit switching and packet switching −

Circuit Switching	**Packet Switching**
Circuit switching requires a dedicated path before sending data from source to	Packet switching does not require any dedicated path to send data from source to

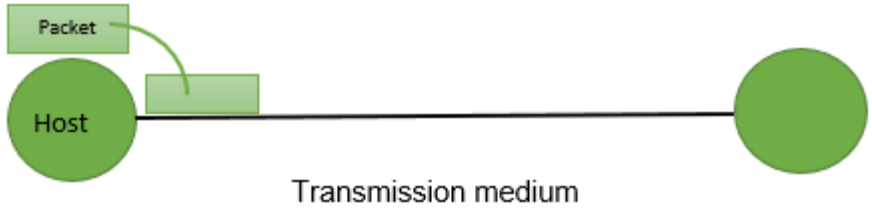
Circuit Switching	Packet Switching
destination.	destination.
It reserves the entire bandwidth in advance.	It does not reserve bandwidth in advance
No store and forward transmission	It supports store and forward transmission
Each packet follows the same route	A packet can follow any route
Call setup is required	No call setup is required
Bandwidth wastage	No bandwidth wastage

6. Discover Different types of Nodal Delay in Detail

The delays, here, means the time for which the processing of a particular packet takes place. We have the following types of delays in computer networks:

1. Transmission Delay:

The time taken to transmit a packet from the host to the transmission medium is called Transmission delay.



For example, if bandwidth is 1 bps (every second 1 bit can be transmitted onto the transmission medium) and data size is 20 bits then what is the transmission delay? If in one second, 1 bit can be transmitted. To transmit 20 bits, 20 seconds would be required.

Let B bps is the bandwidth and L bit is the size of the data then transmission delay is,

Tt = L/B

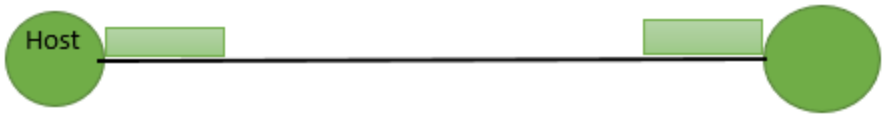
This delay depends upon the following factors:

- If there are multiple active sessions, the delay will become significant.
- Increasing bandwidth decreases transmission delay.

- MAC protocol largely influences the delay if the link is shared among multiple devices.
- Sending and receiving a packet involves a context switch in the operating system, which takes a finite time.

2. Propagation delay:

After the packet is transmitted to the transmission medium, it has to go through the medium to reach the destination. Hence the time taken by the last bit of the packet to reach the destination is called propagation delay.



Factors affecting propagation delay:

1. **Distance –** It takes more time to reach the destination if the distance of the medium is longer.
2. **Velocity –** If the velocity(speed) of the signal is higher, the packet will be received faster.

Tp = Distance / Velocity

Note:

Velocity =3 X 10^8 m/s (for air)

Velocity= 2.1 X 10^8 m/s (for optical fibre)

3. Queueing delay:

Let the packet is received by the destination, the packet will not be processed by the destination immediately. It has to wait in a queue in something called a buffer. So the amount of time it waits in queue before being processed is called queueing delay.

In general, we can't calculate queueing delay because we don't have any formula for that.

This delay depends upon the following factors:

- If the size of the queue is large, the queuing delay will be huge. If the queue is empty there will be less or no delay.
- If more packets are arriving in a short or no time interval, queuing delay will be large.
- The less the number of servers/links, the greater is the queuing delay.

4. Processing delay:

Now the packet will be taken for the processing which is called processing delay.

Time is taken to process the data packet by the processor that is the time required by intermediate routers to decide where to forward the packet, update TTL, perform header checksum calculations.

It also doesn’t have any formula since it depends upon the speed of the processor and the speed of the processor varies from computer to computer.

Note: Both queueing delay and processing delay doesn’t have any formula because they depend on the speed of the processor

This delay depends upon the following factors:

- It depends on the speed of the processor.

Ttotal = Tt + Tp + Tq + Tpro

Ttotal = Tt+Tp

(when taking Tq and Tpro equals to 0)

Unit 2. The Application Layer

1. Define persistent and Non persistent HTTP in detail.

Non-persistent and **persistent** are the two types of HTTP connections used to connect the client with the webserver. The non-persistent connection has connection type 1.0, while the persistent connection has connection type 1.1.

Non-persistent

The non-persistent connection takes a total time of 2RTT + *file transmission time*. It takes the first RTT (round-trip time) to establish the connection between the server and the client. The second RTT is taken to request and return the object. This case stands for a single object transmission.

After the client receives the object in non-persistent, the connection is immediately closed. This is the basic difference between persistent and non-persistent. The persistent connection ensures the transfer of multiple objects over a single connection.

Persistent

A persistent connection takes 1 RTT for the connection and then transfers as many objects, as wanted, over this single connection.

RTT stands for the round-trip time taken for an object request and then its retrieval. In other words, it is the time taken to request the object from the client to the server and then retrieve it from the server back to the client.

Sample problem

Suppose 10 images need to be downloaded from the HTTP server. The total time taken to request and download 10 images in a non-persistent and persistent connection is:

Non-persistent

2 RTT (Connection time) + 2 * 10 RTT= 22 RTT22*RTT*

Persistent

2 RTT (Connection time) + 10 RTT= 12 RTT12*RTT*

2. Describe need of conditional GET in HTTP Messages.

Web caching is done by a Proxy Server – an intermediate entity between the original server and the client. When a client requests for some information (through an HTTP message), it goes through the proxy server, which –

- First checks if it has the copy locally stored.

- If it has, then it forwards the result directly to the client.
- Otherwise, it queries on behalf of the end host, stores a copy of the result locally, and forwards the result back to the end host.

Web caches (or) Proxy Servers are usually installed by ISPs (Internet service providers), Universities, or even Corporate Offices, wherein multiple end hosts are connected to the proxy server.

Installing a proxy server has multiple advantages –

1. It can substantially reduce the response time for repeated requests. (Especially if the bottleneck between the original server and receiver is less than bottleneck between the proxy server and receiver.)
2. It reduces the access link bandwidth (of the university or the office), thereby reducing the cost.
3. It reduces traffic on the Internet as a whole.

But there is one problem.
What if the content was modified on the original server, rendering the copy on the proxy server to be an outdated one?

This is where Conditional GET statements kicks in. When a Proxy server receives an HTTP request, and it has the result stored locally, it still queries the original server, asking if that particular object was modified since the last time it was requested by the proxy server.

The “Conditional GET” statement has an additional field than a normal GET statement, called the “If-modified-since” field, which specifies the last time when the same request was made. The original server either –

- Tells the proxy server that the content was not modified – HTTP 304 status code, or
- Sends the updated content (in case there was some modification done) – HTTP 200 response-message code

If the Proxy server gets a 304 – “No Modification” message, it forwards its local copy to the client. If modification had been there, the Cache forwards the new object, whilst storing it locally along with the date and time it received the new object (so that it can ask the original server later for modifications). For obvious reasons, an HTTP 304 message does not contain a message body.

3. Explain SMTP Protocol. Give comparison of SMTP, POP3 and IMAP.

You must have heard about SMTP, POP3, and IMAP quite often. They are TCP/IP protocols that have been in use for email delivery since 1981. Popular Services like Gmail, Outlook, and Yahoo use this underlying technology to transact over 300 Billion emails every day.

Since you are reading this online, chances are you've already used these three one way or another. Now before we set out to learn what SMTP, IMAP, and POP3 are, let's start with some basics

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What Is A Protocol?

A protocol is a set of standard rules that let electronic devices communicate with each other. Two devices supporting the same protocol can communicate effectively, no matter who their manufacturer is and also what type of devices they are.

TCP/IP stands for "Transmission Control Protocol/Internet Protocol". TCP/IP protocols aim to allow computers to communicate with each other over long-distance networks.

How Are Emails Transferred?

The following three parties are involved in transferring an email:

1. The sender

2. The recipient

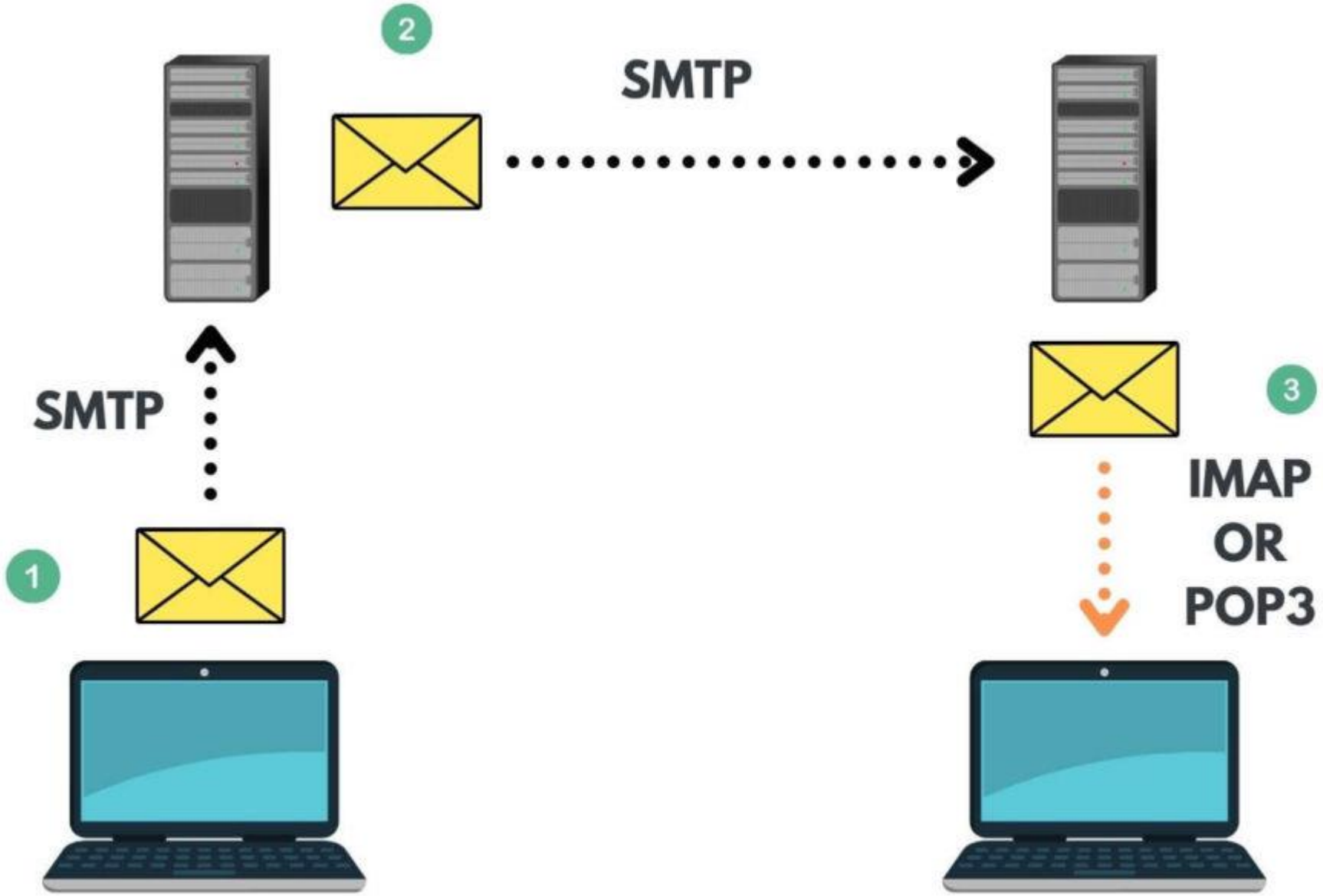
3. A mail server

The email goes to the mail server from the sender, which allows the recipient to receive the email.

Now, to make a connection between the three parties, you need email protocols. SMTP, IMAP, and POP3 are precisely that. They are also the most commonly used TCP/IP protocols.

Journey Of An Email From A Sender To A Recipient

If the sender is jeff@amazon.com and the recipient is elon@spacex.com, following is how an email from the sender reaches the recipient,


2
SMTP
SMTP
3
IMAP
OR
POP3
1

1. The email client of the sender connects to that of the SMTP server (for example, smtp.gmail.com)

2. The SMTP server authenticates the email address of the recipient.

3. After the authorization of the recipient, the Gmail SMTP server sends the email to spacex.com's SMTP server.

4. The SMTP server of spacex.com checks whether elon@spacex.com is valid or not.

5. After that, the SMTP server sends the email to the IMAP/POP3 server.

What Are SMTP, IMAP, And POP3?

What Is SMTP?

SMTP stands for Simple Mail Transfer Protocol. It is the standard protocol for sending email messages using SMTP.

Your email client and internet servers use SMTP after you finish typing your email, upon hitting send. It moves your message over the internet and also makes it land into the recipient's mailbox.

SMTP servers are of two types: Relays and Receivers.

Relays accept the user's emails and route them to the recipient, while receivers deliver them to the mailbox after accepting the email from the relay servers.

Working of SMTP – It involves the SMTP client sending commands and the SMTP server replying to them. The conversation has the following three stages –

1. SMTP handshake – Here, the SMTP client connects to the SMTP server

2. Email transfer – Launches the email transfer

3. Termination – The client and server bid goodbye to each other.

What Is IMAP?

IMAP is an abbreviation for Internet Message Access Protocol. It is a popular protocol for receiving email messages. The most significant advantage of the IMAP protocol is that it lets you receive email on more than one computer (or device). That is because after getting delivered, the email stays on the mail server. In short, you can read your email on your office computer, the desktop at home, your tablet and your smartphone.

Also, it doesn't download the entire message until you open it. That helps in a faster initial connection and startup.

However, it won't perform well when you have a slow internet connection.. And you won't be able to read your emails.

Working of IMAP

The basic IMAP client and server interaction is as mentioned below,

1. The email client of the recipient connects with the server where the message is stored.

2. The recipient can view the message headers on the server.

3. If the recipient selects a message to read it, IMAP downloads that one for the recipient.

What Is POP3?

Like IMAP, POP3 is another protocol for receiving emails. POP3 is an abbreviation for Post Office Protocol, whereas '3' refers to the version. Version 3 is the latest and most widely used one.

It is a simple protocol not having much to offer apart from the download.

POP3 downloads an email from the server to one computer (or device) and ends up removing it from the server once it gets downloaded on your computer. That means it won't be possible to read your email messages from multiple locations, which is not ideal.

The advantages of this protocol are that you can read your emails when you are offline. And the use of the email server's storage is less.

However, as it works with one device and doesn't store the messages on the server, there is a need to backup your computer. Failing to do so will result in you losing all your emails when your computer dies.

Working of POP3

There are four stages in a POP3 connection,

1. Authorization stage – The client connects with the server.

2. Transaction stage – The client retrieves the email

3. Update Stage – The server deletes the message stored

4. The client and server disconnect from each other

Difference Between SMTP, IMAP, And POP3

SMTP VS IMAP

IMAP VS SMTP



IMAP IS USED TO RETRIEVE MESSAGES

SMTP IS FOR SENDING DATA

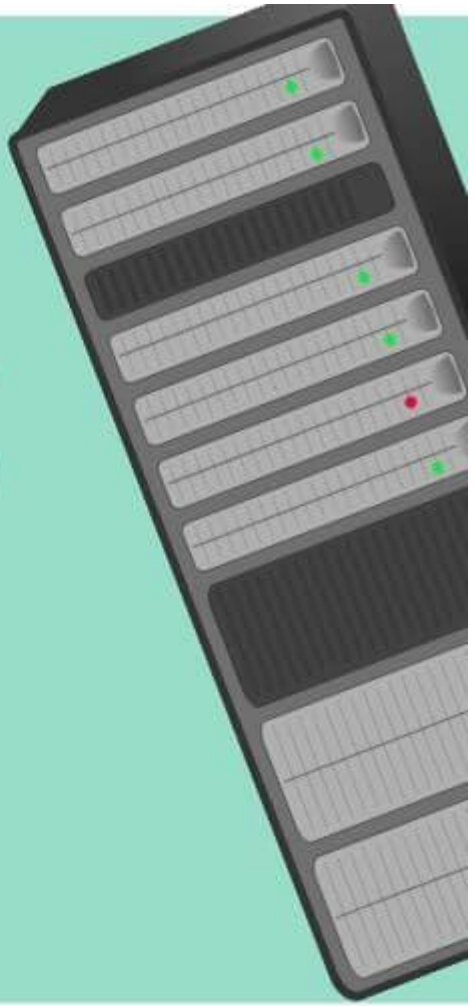
IMAP is used to retrieve messages, while SMTP is for sending them.

IMAP works between the server and client for communication, whereas SMTP works between servers to transfer information. The former is a message transfer agent between user and server, and the latter is a message transfer agent between servers. Read the difference between SMTP VS IMAP.

With IMAP, users can also organize emails onto the server. With SMTP, you can organize emails on client storage.

SMTP vs POP3

SMTP VS POP3



SMTP IS AN MTA OR MESSAGE TRANSFER AGENT FOR SENDING THE MESSAGE TO THE RECIPIENT

POP3 IS AN MAA OR MESSAGE ACCESS AGENT TO ACCESS MESSAGES FROM THE MAILBOX

SMTP is used to send messages, and POP3 is used to access them.

SMTP is an MTA or Message Transfer Agent for sending the message to the recipient. There are two MTAs, namely client MTA and Server MTA. On the other hand, POP3 is an MAA or Message Access Agent to access messages from the mailbox. There are two MAAs as well, namely client MAA and server MAA.

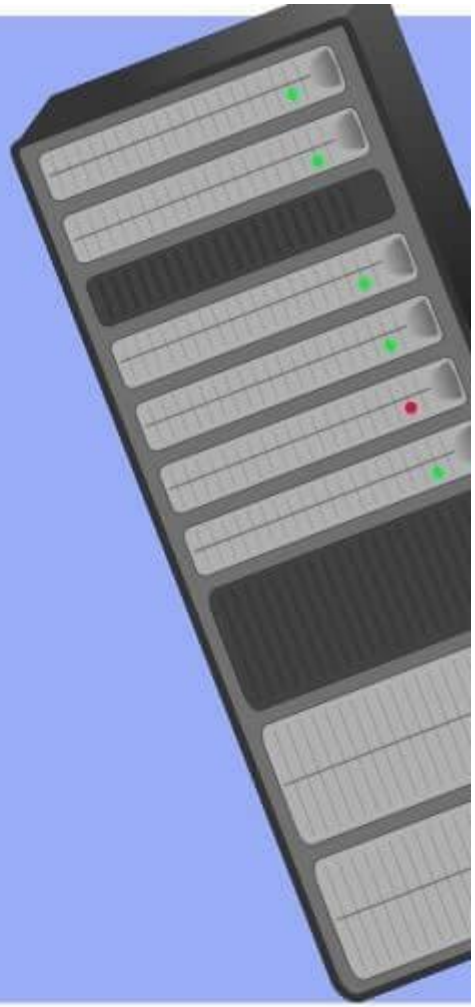
SMTP is referred to as PUSH protocol, and POP3 is called a POP protocol.

SMTP sends the email from the device of the sender to the mailbox from the mail server of the receiver. POP3 lets you retrieve and organize emails from the mailbox on the mail server of the receiver to the computer of the receiver. Read the difference between SMTP VS POP3.

SMTP functions between the mail servers of the sender and receiver, while POP3 functions between the receiver and the mail server of the receiver.

IMAP VS POP3

POP3 VS IMAP



POP3 DOWNLOADS AN EMAIL FROM THE SERVER TO ONE COMPUTER AND THEN DELETES IT FROM THE SERVER

IMAP STORES THE EMAIL ON THE SERVER AND SYNCS IT ACROSS SEVERAL DEVICES TO ACCESS OVER MULTIPLE CHANNELS

Ideally, both IMAP and POP3 are for receiving emails. So, what is the difference between the two protocols?

As already seen above, POP3 downloads an email from the server to one computer and then deletes it from the server. On the contrary, IMAP stores the email on the server and syncs it across several devices. This way, you can access emails across multiple devices. Read the difference between IMAP VS POP3.

Both IMAP and POP3 have spam and virus filters. However, IMAP is more advanced when compared to POP3.

With POP3, you can't organize emails in the mailbox of the mail server, whereas IMAP allows you to organize them.

As a user, you can't create, delete or rename an email on the mail server but it is possible with IMAP. You can download all the emails at once with POP3. While with IMAP, you see the message header before you download the email.

There are two modes of POP3 – delete mode and keep mode.

In the delete mode, your email gets deleted from the mailbox after you retrieve it.

In the keep mode, the email stays in the mailbox even after you retrieve it.

With IMAP, there are several copies of the email on the mail server. That means you can easily retrieve an email message if you lose it.

Which Protocol Is Better To Use – IMAP And POP3?

Choosing between IMAP and POP3 requires you to weigh the advantages and disadvantages of both protocols and consider your requirements.

You should choose IMAP if you want to access mail from multiple devices and organize your email in folders. Also, with IMAP, you will have no server storage space crunch. This protocol is well-suited for a stable internet connection and works well when you want to access your email fast.

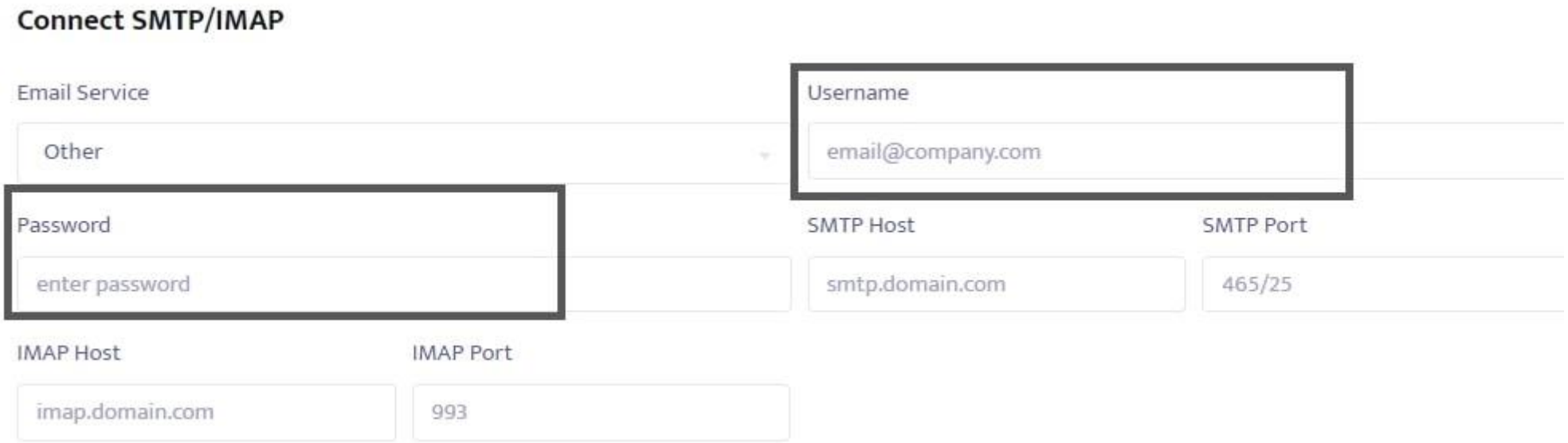
On the other hand, POP3 is the right choice for you when you have an unstable internet connection. If you are concerned about your privacy, POP3 is the best protocol for you as there will be no copies of the email you received on the server. The protocol is well-suited if you are the only one accessing it and also are using just one device for the purpose.

How SalesBlink Uses SMTP And IMAP?

As you have already seen, SMTP is for sending emails, and IMAP is for retrieving them. These two have an important role to play in automating cold email sequences using our software.

For instance, you can’t schedule too many emails on your email service provider, so SalesBlink helps you out with it. You can schedule your email campaign using our software and can schedule your campaign.

To do so, you have to enter your details, such as your username and password, along with the host and port of SMTP and IMAP. Here's a screenshot for your reference.



Next up, SalesBlink uses IMAP to find out whether the outreach emails have got a reply from the prospect or not.

In addition to that, it has a prominent role in enabling you to glance at the history of every prospect and manage them in the CRM. It allows you to track the progress of each prospect of your campaign easily as you can see all the activities on one page.

Why Does SalesBlink Prefer IMAP Over POP3?

SalesBlink is a sales outreach automation suite that uses IMAP instead of POP3 as the former is more effective and helps in serving the purpose.

IMAP stores the message on a server and synchronizes the message across multiple devices while POP3 downloads the email from a server to a single computer, then deletes the email from the server. Therefore, IMAP is more effective for our platform than POP3.

Do You Know The 3 Main Protocols Better Now?

We have seen that the three main TCP/IP protocols for email delivery are SMTP, POP3, and IMAP. While SMTP is for sending emails, IMAP and POP3 are for receiving them. Basically, you can choose between IMAP and POP3 based on your needs. Both have their advantages; however, usually, IMAP suits most people.

FAQs

What is SMTP?

SMTP stands for Simple Mail Transfer Protocol. It is the standard protocol for sending email messages. Your email client and internet servers use SMTP after you finish typing your email, upon hitting send.

What Is IMAP?

IMAP is an abbreviation for Internet Message Access Protocol. It is a popular protocol for receiving email messages. The biggest advantage of the IMAP protocol is that it lets you receive email on more than one computer.

What Is POP3?

POP3 is an abbreviation for Post Office Protocol, whereas '3' refers to the version. Version 3 is the latest and most widely used one. It is a simple protocol not having much to offer apart from the download.

4. Classify different types DNS Services with proper example.

DNS servers play a wide variety of roles—a single name server may be a master for some zones, a slave for others, and provide caching or forwarding services for still others.

The role of the name server is controlled by its configuration file, which in the case of BIND is called **named.conf**. The combination of global parameters in the named.conf file (defined in an options clause) and the zones being serviced (defined in one or more zone clauses) determine the complete functionality of the name server. Depending on the requirements, such configurations can become very complex.

1. Root Servers

Root servers are positioned at the top or root of the DNS hierarchy and maintain data about each of the top-level zones. The root servers are maintained by the NIC and have been moved to a common domain

for consistent naming purposes. The root servers are named as A.root-servers.net., B.root-servers.net., and so on.

2. Primary (Master) Servers

- Each domain must have a primary server. Primary server has the following features.
- There is generally only one primary server per domain.
- They are the system where all the changes are made to the domain.
- They are the authoritative for all domains they serve.
- They periodically update and synchronize secondary servers of the domain.
- In current versions of BIND, they are defined by the type master argument to the zone statement in the configuration file **/etc/named.conf**.

3. Secondary servers

Each domain should have at least one secondary server. In fact,the NIC will not allow a domain to become officially registered as a subdomain of a top-level domain until a site demonstrates two working DNS servers. Secondary servers have the following features.

- There is one or more secondary server per domain.

- They obtain copy of the domain information for all domains they serve from the appropriate primary server or another secondary server for the domain.
- They are authoritative for all the domains they serve.
- They periodically receive updates from the primary servers of the domain.
- They provide load sharing with the primary servers and other servers of the domain.
- They provide redundancy in case one or more other servers are temporarily unavailable.
- They provide more local access to name resolution if placed appropriately.
- In current versions of BIND, they are defined by the type slave argument to the zone statement in the **/etc/named.conf** file.

4. Caching-Only servers

These servers only cache information for any DNS domain. They are not authoritative for any domain. Caching-only servers provide the following features.

- They provide local cache of looked up names.
- They have lower administrative overhead.
- They are never authoritative for any domain.

- They reduce overhead associated with secondary servers performing zone transfers from primary servers.
- They allow DNS client access to local cached naming information without the expense of setting up a DNS primary or secondary server.

5. Forwarding servers

Forwarding servers are a variation on a primary or secondary server and act as focal points for all off-site DNS queries. Designating a server as a forwarding server causes all off-site requests to go through that server first. Forwarding servers have the following features.

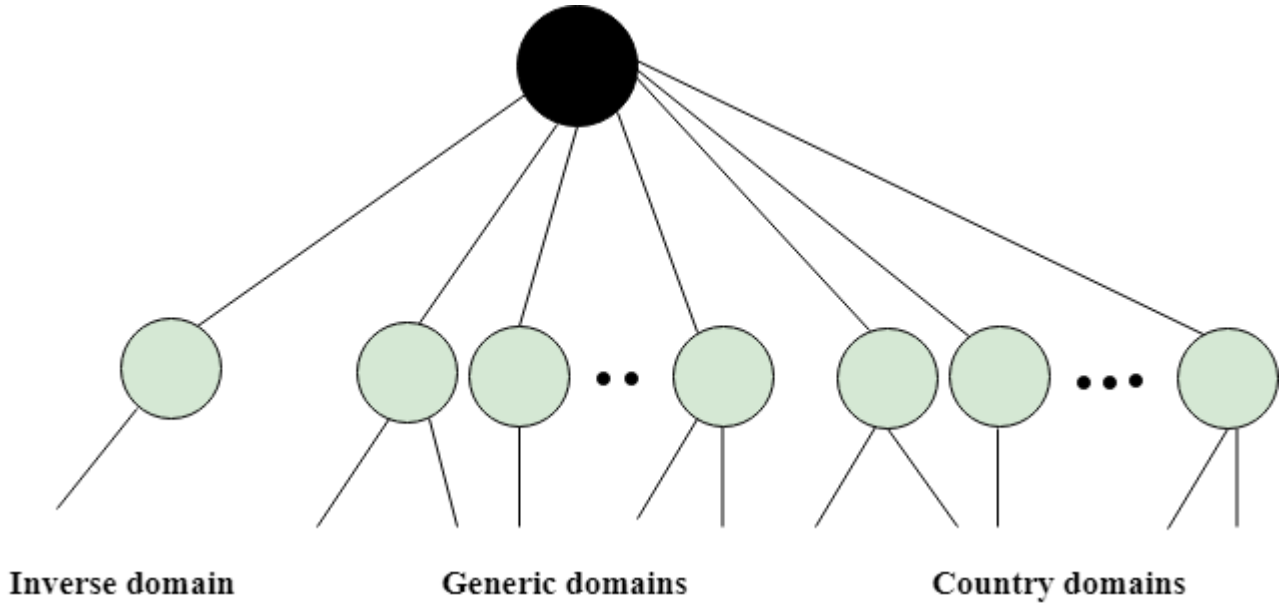
- They are used to centralize off-site requests.
- The server being used as a forwarder builds up a rich cache of information.
- All off-site queries go through forwarders first.
- They reduce the number of redundant off-site requests.
- No special setup on forwarders is required.
- If forwarders fail to respond to queries, the local server can still contact a remote site, DNS servers itself.

- 5. Prepare a short note on DNS Database.

An application layer protocol defines how the application processes running on different systems, pass the messages to each other.

- DNS stands for Domain Name System.
- DNS is a directory service that provides a mapping between the name of a host on the network and its numerical address.
- DNS is required for the functioning of the internet.
- Each node in a tree has a domain name, and a full domain name is a sequence of symbols specified by dots.
- DNS is a service that translates the domain name into IP addresses. This allows the users of networks to utilize user-friendly names when looking for other hosts instead of remembering the IP addresses.
- For example, suppose the FTP site at EduSoft had an IP address of 132.147.165.50, most people would reach this site by specifying ftp.EduSoft.com. Therefore, the domain name is more reliable than IP address.

DNS is a TCP/IP protocol used on different platforms. The domain name space is divided into three different sections: generic domains, country domains, and inverse domain.



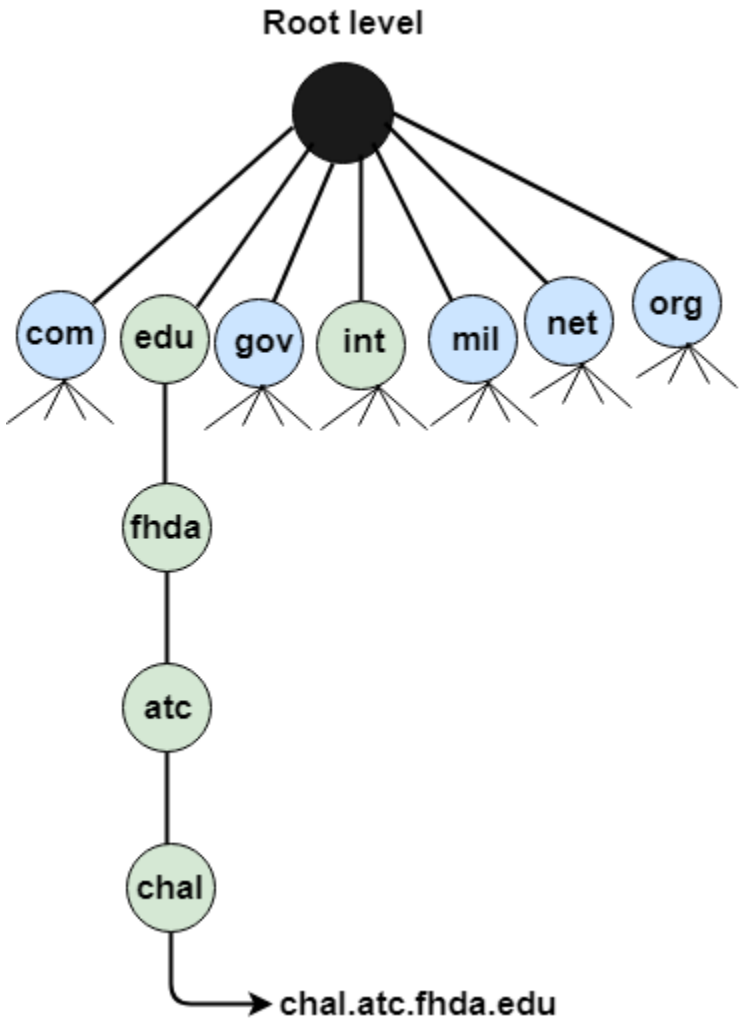
Generic Domains

- It defines the registered hosts according to their generic behavior.
- Each node in a tree defines the domain name, which is an index to the DNS database.
- It uses three-character labels, and these labels describe the organization type.

Label	Description

aero	Airlines and aerospace companies
biz	Businesses or firms
com	Commercial Organizations
coop	Cooperative business Organizations
edu	Educational institutions
gov	Government institutions
info	Information service providers
int	International Organizations
mil	Military groups

museum	Museum & other nonprofit organizations
name	Personal names
net	Network Support centers
org	Nonprofit Organizations
pro	Professional individual Organizations



Country Domain

The format of country domain is same as a generic domain, but it uses two-character country abbreviations (e.g., us for the United States) in place of three character organizational abbreviations.

Inverse Domain

The inverse domain is used for mapping an address to a name. When the server has received a request from the client, and the server contains the files of only authorized clients. To determine whether the client is on the authorized list or not, it sends a query to the DNS server and ask for mapping an address to the name.

Working of DNS

- DNS is a client/server network communication protocol. DNS clients send requests to the. server while DNS servers send responses to the client.
- Client requests contain a name which is converted into an IP address known as a forward DNS lookups while requests containing an IP address which is converted into a name known as reverse DNS lookups.
- DNS implements a distributed database to store the name of all the hosts available on the internet.
- If a client like a web browser sends a request containing a hostname, then a piece of software such as **DNS resolver** sends a request to the DNS server to obtain the IP address of a hostname. If DNS server does not contain the IP address associated with a hostname, then it forwards the request to

another DNS server. If IP address has arrived at the resolver, which in turn completes the request over the internet protocol.

6. Discuss the concept of Cookies and its components with suitable example.

Cookies are small files of information that a web server generates and sends to a web browser. Web browsers store the cookies they receive for a predetermined period of time, or for the length of a user's session on a website. They attach the relevant cookies to any future requests the user makes of the web server.

Cookies help inform websites about the user, enabling the websites to personalize the user experience. For example, ecommerce websites use cookies to know what merchandise users have placed in their shopping carts. In addition, some cookies are necessary for security purposes, such as authentication cookies (see below).

The cookies that are used on the Internet are also called "HTTP cookies." Like much of the web, cookies are sent using the HTTP protocol.

Where are cookies stored?

Web browsers store cookies in a designated file on users' devices. The Google Chrome web browser, for instance, stores all cookies in a file labeled "Cookies." Chrome users can view the cookies stored by the browser by opening developer tools, clicking the "Application" tab, and clicking on "Cookies" in the left side menu.

What are cookies used for?

User sessions: Cookies help associate website activity with a specific user. A session cookie contains a unique string (a combination of letters and numbers) that matches a user session with relevant data and content for that user.

Suppose Alice has an account on a shopping website. She logs into her account from the website's homepage. When she logs in, the website's server generates a session cookie and sends the cookie to Alice's browser. This cookie tells the website to load Alice's account content, so that the homepage now reads, "Welcome, Alice."

Alice then clicks to a product page displaying a pair of jeans. When Alice's web browser sends an HTTP request to the website for the jeans product page, it includes Alice's session cookie with the request. Because the website has this cookie, it recognizes the user as Alice, and she does not have to log in again when the new page loads.

Personalization: Cookies help a website "remember" user actions or user preferences, enabling the website to customize the user's experience.

If Alice logs out of the shopping website, her username can be stored in a cookie and sent to her web browser. Next time she loads that website, the web browser sends this cookie to the web server, which then prompts Alice to log in with the username she used last time.

Tracking: Some cookies record what websites users visit. This information is sent to the server that originated the cookie the next time the browser has to load content from that server. With third-party tracking cookies, this process takes place anytime the browser loads a website that uses that tracking service.

If Alice has previously visited a website that sent her browser a tracking cookie, this cookie may record that Alice is now viewing a product page for jeans. The next time Alice loads a website that uses this tracking service, she may see ads for jeans.

However, advertising is not the only use for tracking cookies. Many analytics services also use tracking cookies to anonymously record user activity. (Cloudflare Web Analytics is one of the few services that does not use cookies to provide analytics, helping to protect user privacy.)

What are the different types of cookies?

Some of the most important types of cookies to know include:

Session cookies

A session cookie helps a website track a user's session. Session cookies are deleted after a user's session ends — once they log out of their account on a website or exit the website. Session cookies have no expiration date, which signifies to the browser that they should be deleted once the session is over.

Persistent cookies

Unlike session cookies, persistent cookies remain in a user's browser for a predetermined length of time, which could be a day, a week, several months, or even years. Persistent cookies always contain an expiration date.

Authentication cookies

Authentication cookies help manage user sessions; they are generated when a user logs into an account via their browser. They ensure that sensitive information is delivered to the correct user sessions by associating user account information with a cookie identifier string.

Tracking cookies

Tracking cookies are generated by tracking services. They record user activity, and browsers send this record to the associated tracking service the next time they load a website that uses that tracking service.

Zombie cookies

Like the "zombies" of popular fiction, zombie cookies regenerate after they are deleted. Zombie cookies create backup versions of themselves outside of a browser's typical cookie storage location. They use these backups to reappear within a browser after they are deleted. Zombie cookies are sometimes used by unscrupulous ad networks, and even by cyber attackers.

What is a third-party cookie?

A third-party cookie is a cookie that belongs to a domain other than the one displayed in the browser. Third-party cookies are most often used for tracking purposes. They contrast with first-party cookies, which are associated with the same domain that appears in the user's browser.
When Alice does her shopping at jeans.example.com, the jeans.example.com origin server uses a session cookie to remember that she has logged into her account. This is an example of a first-party cookie. However, Alice may not be aware that a cookie from example.ad-network.com is also stored in her browser and is tracking her activity on jeans.example.com, even though she is not currently accessing example.ad-network.com. This is an example of a third-party cookie.

How do cookies affect user privacy?

As described above, cookies can be used to record browsing activity, including for advertising purposes. However, many users do not want their online behavior to be tracked. Users also lack visibility or control over what tracking services do with the data they collect.
Even when cookie-based tracking is not tied to a specific user's name or device, with some types of tracking it could still be possible to link a record of a user's browsing activity with their real identity. This information could be used in any number of ways, from unwanted advertising to the monitoring, stalking, or harassment of users. (This is not the case with all cookie usage.)
Some privacy laws, like the EU's ePrivacy Directive, address and govern the use of cookies. Under this directive, users have to provide "informed consent" — they have to be notified of how the website uses cookies and agree to this usage — before the website can use cookies. (The exception to this is cookies that are "strictly necessary" for the website to function.) The EU's General Data Protection Regulation (GDPR) considers cookie identifiers to be personal data, so its rules apply to cookie usage in the EU as well. Also, any personal data collected by cookies falls under the GDPR's jurisdiction.

Largely because of these laws, many websites now display cookie banners that allow users to review and control the cookies those websites use.

7. Discuss the high-level view of Internet e-mail system and its major components

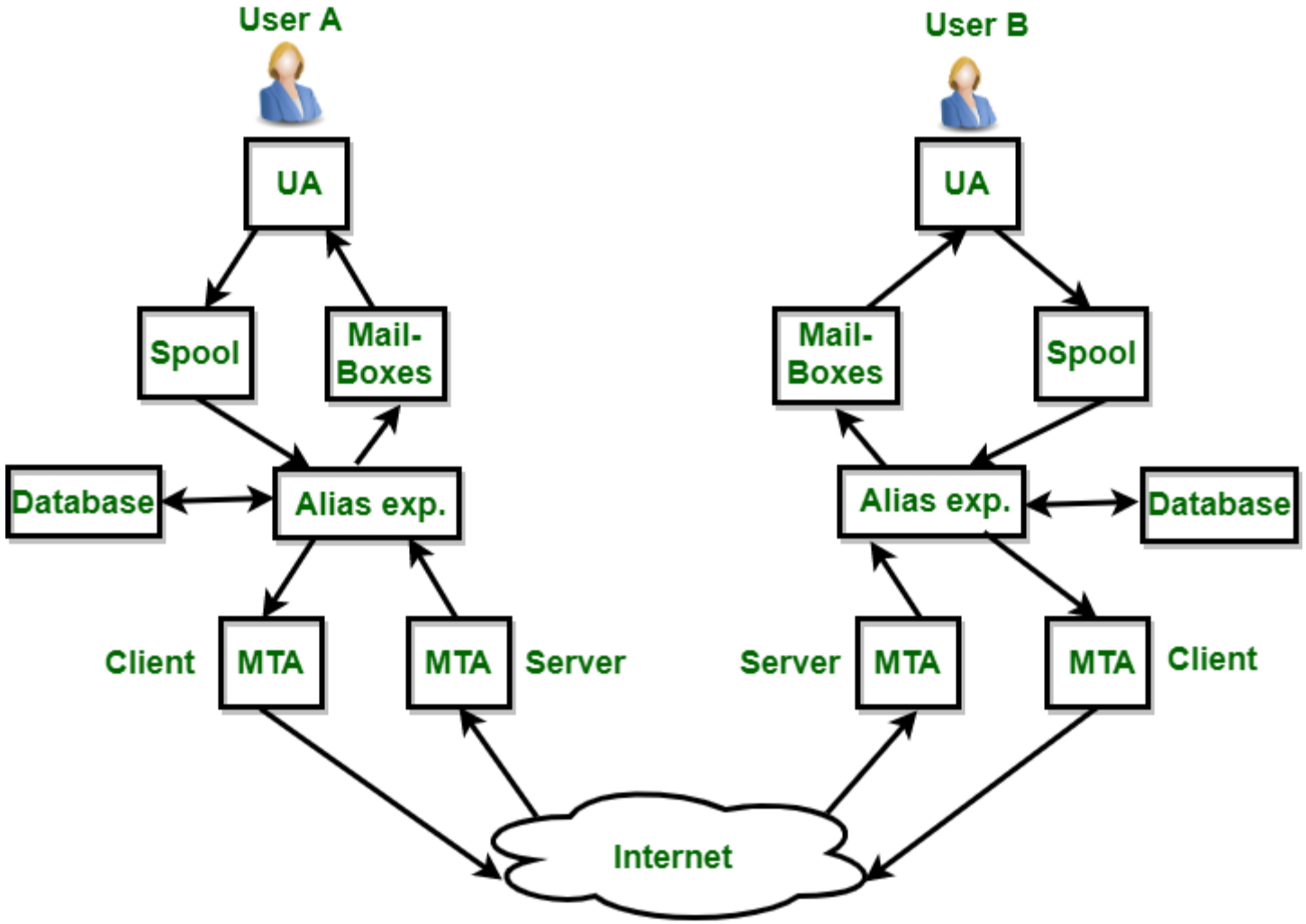
Electronic Mail (e-mail) is one of most widely used services of Internet. This service allows an Internet user to send a **message in formatted manner (mail)** to the other Internet user in any part of world. Message in mail not only contain text, but it also contains images, audio and videos data. The person who is sending mail is called **sender** and person who receives mail is called **recipient**. It is just like postal mail service.

Components of E-Mail System :

The basic components of an email system are : User Agent (UA), Message Transfer Agent (MTA), Mail Box, and Spool file. These are explained as following below.

1. **User Agent (UA) :**
   The UA is normally a program which is used to send and receive mail. Sometimes, it is called as mail reader. It accepts variety of commands for composing, receiving and replying to messages as well as for manipulation of the mailboxes.

2. **Message Transfer Agent (MTA) :**
   MTA is actually responsible for transfer of mail from one system to another. To send a mail, a system must have client MTA and system MTA. It transfer mail to mailboxes of recipients if they are connected in the same machine. It delivers mail to peer MTA if destination mailbox is in another machine. The delivery from one MTA to another MTA is done by Simple Mail Transfer Protocol.



3. **Mailbox :**
   It is a file on local hard drive to collect mails. Delivered mails are present in this file. The user can read

it delete it according to his/her requirement. To use e-mail system each user must have a mailbox . Access to mailbox is only to owner of mailbox.

4. **Spool file :**
   This file contains mails that are to be sent. User agent appends outgoing mails in this file using SMTP. MTA extracts pending mail from spool file for their delivery. E-mail allows one name, an **alias**, to represent several different e-mail addresses. It is known as **mailing list**, Whenever user have to sent a message, system checks recipients's name against alias database. If mailing list is present for defined alias, separate messages, one for each entry in the list, must be prepared and handed to MTA. If for defined alias, there is no such mailing list is present, name itself becomes naming address and a single message is delivered to mail transfer entity.

Services provided by E-mail system :

- **Composition –**
  The composition refer to process that creates messages and answers. For composition any kind of text editor can be used.
- **Transfer –**
  Transfer means sending procedure of mail i.e. from the sender to recipient.
- **Reporting –**
  Reporting refers to confirmation for delivery of mail. It help user to check whether their mail is delivered, lost or rejected.
- **Displaying –**
  It refers to present mail in form that is understand by the user.

- **Disposition –**
  This step concern with recipient that what will recipient do after receiving mail i.e save mail, delete before reading or delete after reading.